S_{plan}: Compact Learned Symbolic Planning Tokens for Music-Aware Audio Language Models

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Abstract

Audio language models have achieved strong performance across diverse audio tasks, yet music understanding and generation remain challenging because the structural properties of music—harmonic progressions, rhythmic regularity, instrument co-occurrence—are encoded in MIDI symbolic representations rather than acoustic waveforms. We introduce S_{plan}, a *compact learned symbolic planning token* that compresses MIDI-derived musical structure into a fixed-length residual vector quantization (RVQ) stream suitable for integration into unified audio language models. Rather than appending raw MIDI tokens—which suffer from sequence-length brittleness and alignment artifacts—S_{plan} learns a low-rate bottleneck using frame-level pitch, rhythm, and timbral features from the Slakh paired audio–MIDI corpus. We evaluate S_{plan} through a systematic experiment ladder with aligned versus shuffled versus dummy controls at each stage. Our strongest result (EXP034B) is a complete four-condition open-gate matched comparison where aligned S_{plan} achieves test cross-entropy of 5.600, compared to 5.614 for both alignment-corrupted controls and 5.859 for the no-symbolic baseline—a clear positive result. In parallel, MIR attribute probing (EXP031A) shows that pitch-class macro-F1 (0.854), segment density accuracy (0.874), and weak key accuracy (0.557) are recoverable from S_{plan} embeddings, while meter and instrument micro-F1 are substantially explained by dataset priors. We argue that controlled evaluation with matched corruptions is essential for credible symbolic-fusion claims, and establish the conservative but supported claim that S_{plan} is a compact symbolic codec with acoustic-token utility and music-structure content under controlled Slakh experiments.

1 Introduction

Unified audio language models (LALMs) (Yang et al., 2024, 2026; Rubenstein et al., 2023; Borsos et al., 2023) have demonstrated strong performance on diverse tasks by predicting discrete audio tokens autoregressively. Music understanding and generation, however, pose a persistent challenge: the structural properties most salient to musicians and listeners—harmonic progressions, rhythmic regularity, instrument co-occurrence, phrase boundaries—are encoded in MIDI symbolic representations rather than waveform statistics. While MIDI-to-audio synthesis pipelines use explicit symbolic intermediates, integrating these representations into LALMs has remained an open problem.

The most direct approach is to append raw MIDI tokens to the audio context. This approach fails in practice for three reasons. First, standard MIDI tokenizations for 30-second windows routinely produce thousands of tokens, exceeding the context budgets of transformer architectures designed for audio. Second, MIDI token alignment with audio tokens is non-trivial: raw MIDI events are asynchronous with audio frame boundaries, creating positional artifacts. Third, and most critically, naive concatenation does not guarantee that the model *uses* symbolic content causally—as we show in EXP027, aligned and shuffled MIDI insertions produce nearly identical predictions, suggesting the model ignores the symbolic stream entirely.

We address these challenges with S_{plan}, a *learned symbolic planning token* that compresses MIDI-derived musical structure into a compact, fixed-length, multi-stream RVQ representation. S_{plan} is designed as a stream-native addition to the UniAudio 2.0 framework (Yang et al., 2026), which organizes audio into reasoning tokens R_a and semantic acoustic tokens C_a. We insert S_{plan} between R_a and C_a, forming the three-stream sequence $R_a \rightarrow S_{\text{plan}} \rightarrow C_a$ that fits the multi-stream positional encoding of the base model without exceeding context limits.

Our evaluation philosophy treats controlled corruptions as first-class evidence. At every stage we compare aligned S_{plan} against both a shuffled variant (temporal order of symbolic frames permuted) and a dummy variant (zero or random codes), under matched training budgets. This approach, inspired by probing literature in NLP (Hewitt and Manning, 2019; Tenney et al., 2019), lets us separate symbolic content effects from token budget effects.

Contributions. We contribute:

1. A low-rate RVQ symbolic codec (S_{plan}) that compresses MIDI-derived frame features into 4×30 tokens per 30-second window—a $10\times$–$50\times$ reduction compared to raw MIDI tokenizations.

2. A systematic evaluation ladder across four independent experiments (EXP024, EXP025, EXP029, EXP034B) using matched aligned/shuffled/dummy controls, demonstrating positive and consistent symbolic utility.

3. A complete four-condition open-gate hardening experiment (EXP034B) with all conditions finished under identical settings, yielding the clearest causal evidence of alignment-specific symbolic benefit.

4. An axis-level analysis of MIR attribute recovery (EXP031A) that identifies which symbolic attributes are robustly attributable to S_{plan} content versus dataset priors.

5. Honest claim boundaries: we document what each experiment can and cannot support, including two negative results (EXP027, EXP028) that informed our interface design.

2 Related Work

2.1 Audio Language Models

AudioLM (Borsos et al., 2023) pioneered hierarchical discrete token prediction for long-form audio continuation. AudioPaLM (Rubenstein et al., 2023) unified speech and audio tasks under a single LLM. UniAudio (Yang et al., 2024) extended this to a broad task catalog using a multi-stream tokenizer. UniAudio 2.0 (Yang et al., 2026) introduced ReasoningCodec, which factorizes audio into reasoning tokens (high-level, language-aligned) and reconstruction tokens (fine-grained acoustic detail), supervised via SFT and GRPO. UALM (Team, 2025) uses a Qwen2.5-based decoder with continuous audio embeddings and X-codec tokens, demonstrating competitive MMAU performance but without explicit symbolic grounding.

None of these systems incorporate a learned *symbolic* planning stream verified by alignment controls. Our work fills this gap on the music domain.

2.2 Symbolic Music Representations

Symbolic music can be represented via raw MIDI events, piano rolls, or higher-level abstractions (Fradet et al., 2021; Hawthorne et al., 2019). Raw MIDI tokenizations suffer from extreme sequence lengths and are not directly comparable to audio token rates. Compact representations such as chord sequences or beat annotations are hand-designed and limited in expressiveness. S_{plan} learns a compact representation end-to-end from paired MIDI features, avoiding hand-design while keeping sequence length tractable.

2.3 Vector Quantization for Audio

Residual vector quantization (RVQ) has become the standard for neural audio codecs (Zeghidour et al., 2021; Défossez et al., 2023; Kumar et al., 2024). We apply the same quantization principle to symbolic features rather than waveforms, enabling S_{plan} to use the same multi-stream position encoding as the acoustic codec.

2.4 Music Information Retrieval

MIR tasks including key estimation, beat tracking, instrument recognition, and chord detection provide objective, structured evaluation targets (Raffel et al., 2014). We use MIR attributes from Slakh MIDI metadata as supervision and evaluation targets for S_{plan}, yielding interpretable quality measures beyond cross-entropy.

3 Background: UniAudio 2.0

UniAudio 2.0 (Yang et al., 2026) organizes audio into two factorized streams:

Reasoning tokens R_a: Low-rate, language-aligned tokens from a ReasoningCodec encoder, trained with SFT + GRPO on ASR, captioning, and audio-reasoning tasks. These tokens capture high-level semantic and contextual content.

Reconstruction tokens C_a: An 8-codebook RVQ stream for fine-grained acoustic reconstruction and generation, trained after the reasoning branch is frozen.

The unified autoregressive model predicts C_a conditioned on text and R_a, using a single transformer with multi-stream positional encoding. Audio-specific experts update only audio-position hidden states. This factorization allows both understanding (via R_a) and generation (via C_a) within a single framework.

Our work asks: *can a symbolic planning stream S_{plan}, inserted between R_a and C_a, carry music-structural information that R_a alone does not provide?*

4 The S_{plan} System

4.1 MIDI Feature Extraction

Given a 30-second Slakh audio-MIDI window, we extract frame-level features from the aligned MIDI file at 100 ms resolution:

- **Binary features:** Instrument-class presence vector (34 Slakh classes), 12-dim pitch-class binary, beat/downbeat phase indicator.
- **Continuous features:** Instantaneous tempo (BPM), note density (events/second), polyphony count, instrument-weighted energy proxy.

Features are normalized per-dimension across the training split. The final frame vector has 53 dimensions (41 binary, 12 continuous).

4.2 Symbolic RVQ Codec (EXP024)

We train a residual vector quantizer over the frame-level feature sequence with the following architecture:

- **Encoder:** Two-layer 1D CNN followed by temporal pooling that maps variable-length feature sequences to a fixed 30-frame representation per 30-second window.
- **Codebook:** 4 RVQ stages, 512 codes each, trained with exponential moving average updates and codebook restart.
- **Decoder:** Symmetric 1D CNN that reconstructs the original feature sequence from quantized codes.
- **Loss:** Binary cross-entropy for binary feature dimensions plus mean squared error (MSE) for continuous dimensions.

Training runs for 42,000 steps on the Slakh training split.

4.3 Stream-Native Integration

We integrate S_{plan} into the UniAudio 2.0 sequence as a third stream between R_a and C_a:

$$\underbrace{[\text{text}]}_{\text{1 stream}} \rightarrow \underbrace{[R_a]}_{\text{8 streams}} \rightarrow \underbrace{[S_{\text{plan}}]}_{\text{4 streams}} \rightarrow \underbrace{[C_a]}_{\text{8 streams}}$$

The sequence layout produces 151 reason frames, 30 symbolic frames, and 376 semantic frames (total 557 positions), fitting within the 1024 context limit with 0% truncation.

We use LoRA adapters (Hu et al., 2022) on the UniAudio 2.0 transformer attention modules for parameter-efficient fine-tuning (140 modules, $\sim$10.8M trainable parameters in the reason-only configuration and $\sim$253M for symbolic stream variants), with a learned scalar gating mechanism that controls the magnitude of symbolic stream influence.

4.4 Why Not Raw MIDI?

Our design is motivated by three failures of raw MIDI appending (EXP020–023):

- **Length:** Standard MIDITok (Fradet et al., 2021) tokenizations of 30-second windows average $\sim$2,000–8,000 tokens, saturating the 1024 context.
- **Alignment:** Raw MIDI event timestamps are asynchronous with audio frames, creating positional artifacts that confuse multi-stream positional encoding.
- **Interpretability:** A compact, fixed-length S_{plan} stream with known RVQ structure is more amenable to controlled shuffled/dummy comparisons than a variable-length event sequence.

Low-rate rule summaries (hand-crafted text-like symbolic summaries, average $\sim$149 tokens) avoid the length problem but remain hand-designed. EXP025 shows that learned S_{plan} outperforms rule summaries in acoustic-token utility (see §6).

5 Experimental Setup

5.1 Dataset

All experiments use Slakh2100 (Manilow et al., 2019), a multi-track paired audio-MIDI dataset synthesized with professional-grade sample-based virtual instruments. We split at the *track* level before windowing to prevent train-test contamination, yielding:

Split	Tracks	30 s windows	Notes
Train	1,289	11,279	Track-level split
Validation	270	2,334	Zero track overlap
Test	151	1,385	Held-out tracks

Table 1: Slakh dataset statistics. All symbolic comparisons use the same split.

Metric	Value
Validation reconstruction loss	0.00918
Binary feature accuracy	0.99777
Continuous feature MSE	0.00172
Codebook utilization (books 1–4)	31.3% / 71.9% / 80.7% / 81.2%

Table 2: EXP024 validation metrics at step 42,000.

5.2 Control Protocol

For every experiment that uses S_{plan}, we train and evaluate three conditions:

Aligned S_{plan}: The S_{plan} tokens corresponding to the audio window's own MIDI file, preserving temporal and content alignment.

Shuffled S_{plan}: The S_{plan} tokens of a randomly selected *different* window, preserving token statistics but breaking content alignment.

Dummy stream: Zero or uniformly random RVQ codes, providing a token-budget control that tests capacity effects without any symbolic content.

A gain of aligned over *both* shuffled and dummy establishes that alignment-specific content—not merely extra tokens or capacity—drives the improvement.

5.3 Metrics

We use the following evaluation metrics:

- **Cross-entropy (CE) / perplexity (PPL):** Token-level prediction loss on C_a acoustic tokens.
- **Token accuracy:** Exact-match accuracy on held-out C_a tokens.
- **Symbolic gate:** The learned gate value, measuring the fraction of symbolic stream influence admitted to the acoustic prediction path.
- **MIR metrics (EXP031A):** Instrument macro/micro-F1, pitch-class macro-F1, meter accuracy, weak key accuracy, segment density accuracy, and segment polyphony accuracy.

6 Results

6.1 EXP024: Symbolic Codec Reconstruction

The S_{plan} RVQ codec trains stably over 42,000 steps without codebook collapse.

Per-window analysis on three validation fixed examples confirms binary accuracy of 1.000 and continuous MSE in $[0.0013, 0.0025]$, with 15–26 unique codes per codebook book (out of 512), indicating the codec is not collapsed but uses a focused subset of the codebook. The first RVQ book (31.3% utilization) covers the most common frame-level patterns; later books refine residuals.

These results establish that S_{plan} is technically feasible as a symbolic bottleneck: it compresses MIDI-derived structure without collapse, and reconstructs features faithfully.

6.2 EXP025: Acoustic-Token Utility Proxy

EXP025 tests whether a fixed S_{plan} (codec frozen) carries information useful for predicting UniAudio semantic acoustic tokens C_a from reason tokens R_a. We train a small transformer head on top of the fixed representations and compare conditions on held-out test windows.

Condition	Valid CE	Test CE	Test PPL
Reason only	7.5650	7.6573	2116.1
Low-rate rule summary	7.5534	7.6445	2089.2
Aligned S_{plan}	**7.5073**	**7.5963**	**1990.9**
↪ Shuffled S_{plan}	7.5376	7.6307	2060.6
↪ Dummy stream	7.6023	7.6942	2195.5
↪ Random stream	7.5400	7.6325	2064.2

Table 3: EXP025 aggregate results on 1,385 held-out test windows (5,000 training steps, seed 260625).

Aligned S_{plan} achieves the lowest test CE (7.5963), outperforming reason-only (7.6573, $\Delta = -0.061$), low-rate rule summaries (7.6445, $\Delta = -0.048$), shuffled S_{plan} (7.6307, $\Delta = -0.034$), and dummy stream (7.6942, $\Delta = -0.098$). The fact that aligned S_{plan} outperforms all corrupted controls—including the shuffled variant which preserves token statistics but breaks temporal alignment—indicates the improvement is not simply a capacity effect.

6.3 EXP027–028: Negative Results and Interface Lessons

EXP027 (naive insertion): Inserting S_{plan} tokens as additional context rows into the released UniAudio 2.0 LALM interface produced nearly identical results across aligned (5.5113), shuffled (5.5119), and dummy (5.5100) conditions, all worse than reason-only alone (5.3598). This negative result is informative: the model ignored symbolic content when it was injected as an external side-channel incompatible with the multi-stream positional encoding.

EXP028 (LoRA adapter): Adding LoRA adapters plus a tiny learned gate stabilized training (valid CE 5.4116) but the gate converged near zero (3.36×10^{-4}), indicating the symbolic stream was admitted at negligible magnitude. Stability without symbolic use is not fusion evidence.

These negative results motivated the stream-native interface design of EXP029 and EXP034B, which aligns S_{plan} with the multi-stream position encoding expected by the base model.

6.4 EXP029: Stream-Native Interface

EXP029 implements the full three-stream layout $R_a \to S_{\text{plan}} \to C_a$ with all streams aligned to the base model's positional encoding.

Condition	Test CE	Test Acc	Prefix-8 CE	Gate
Aligned S_{plan}	**5.766**	**0.1097**	**5.982**	0.0033
↪ Shuffled S_{plan}	5.785	0.1067	6.000	0.0032
↪ Dummy stream	5.785	0.1068	6.004	0.0034

Table 4: EXP029 aggregate results on held-out test set (3,000 steps, seed 260630). Gate is the learned symbolic contribution magnitude.

Aligned S_{plan} achieves test CE 5.766, compared to shuffled 5.785 ($\Delta = -0.018$) and dummy 5.785 ($\Delta = -0.019$). The margin is modest but consistent: on four validation fixed windows, two show strong positive margins ($+0.054$, $+0.004$) and two show near-zero or slightly negative margins (-0.006, -0.007), characteristic of a small but real signal that has not yet saturated the learning budget.

6.5 EXP034B: Open-Gate Music-Claim Hardening (Primary Result)

EXP034B is our strongest and most rigorous comparison. All four conditions run under identical hyperparameters (800 steps, batch 1, gradient accumulation 16, LoRA lr 10^{-5}, symbolic lr 3×10^{-4}, gate lr 2×10^{-3}, seed 260635), differing only in the symbolic stream variant.

Condition	Test CE	Test PPL	Test Acc	Gate
Reason-only baseline	5.859	350.4	0.1039	0.000
Aligned S_{plan} stream	**5.600**	**270.4**	**0.1207**	0.177
↪ Shuffled S_{plan}	5.614	274.3	0.1191	0.167
↪ Dummy stream	5.614	274.3	0.1193	0.163

Table 5: EXP034B final test metrics at step 800. All four conditions complete (`claim_ready = true`).

Three effects are visible:

Stream capacity effect: All three symbolic stream variants substantially outperform reason-only ($\Delta \approx -0.245$ CE, PPL $350 \to 274$). This reflects capacity and positional structure added by any 30-frame stream in the S_{plan} slot.

Alignment-specific effect: Aligned S_{plan} outperforms both shuffled ($\Delta = -0.014$) and dummy ($\Delta = -0.015$) by a consistent margin, confirming that temporal alignment of symbolic content to the corresponding audio window provides additional benefit beyond token capacity alone.

Gate behavior: The learned gate opens more for aligned (0.177) than shuffled (0.167) or dummy (0.163), suggesting the model learns to admit more information from the aligned stream—consistent with it carrying useful content.

Across EXP025 ($\Delta = -0.034$), EXP029 ($\Delta = -0.018$), and EXP034B ($\Delta = -0.014$), the alignment-specific margin is positive and consistent, providing convergent evidence of symbolic utility across different training regimes and evaluation protocols.

6.6 EXP031A: MIR Structure Recovery

EXP031A evaluates whether MIR attributes can be recovered from S_{plan} embeddings via supervised probing heads, testing whether the codec preserves musically meaningful structure.

A zero-feature control (inputs zeroed, only head bias and BatchNorm statistics available) reveals which axes are prior-dominated:

- **Safe positive axes:** Weak key accuracy (0.557 vs. 0.126, $\Delta = +0.431$) and pitch-class macro-F1 (0.854 vs. 0.830, $\Delta = +0.024$) show the largest gaps attributable to S_{plan} content.
- **Instrument macro-F1:** Some feature-specific signal (0.501 vs. 0.426, $\Delta = +0.075$), but class imbalance and threshold

Metric	Aligned S_{plan} (test / step 625)	Zero-feature (valid / step 500)
Instrument macro-F1	0.501	0.426
Instrument micro-F1	0.791	**0.835**
Pitch-class macro-F1	**0.854**	0.830
Meter accuracy	0.783	**0.855**
Weak key accuracy	**0.557**	0.126
Segment density acc.	**0.874**	0.879
Segment polyphony acc.	**0.761**	–

Table 6: EXP031A MIR recovery at step 625 (aligned S_{plan}) versus zero-feature control at step 500. Bold indicates the row where the feature-dependent model is more robustly attributable. Label-shuffle and target-prior controls are pending.

calibration must be addressed before strong claims.

- **Prior-dominated axes:** Instrument micro-F1 (0.791 vs. 0.835, zero-feature *higher*) and meter accuracy (0.783 vs. 0.855, zero-feature *higher*) are largely explained by common-class and common-meter priors, and should not be headlined as symbolic contributions.

These results demonstrate that S_{plan} preserves musically structured content on pitch and key axes, while confirming that label-shuffle and target-prior controls are necessary before causal claims on instrument and meter axes.

7 Discussion

The gap between capacity and alignment. The dominant effect in EXP034B is the stream capacity effect ($\Delta \approx -0.245$ CE versus reason-only). This arises because any 30-frame stream in the S_{plan} slot, regardless of content, expands the context available for C_a prediction. The alignment-specific effect ($\Delta \approx -0.014$–-0.034 CE across experiments) is a reliable but smaller fraction. This decomposition is important for paper claims: a strong positive result must report both effects, not conflate them.

Why consistency across experiments matters. The alignment-specific gain is positive across EXP025 (cross-prediction proxy), EXP029 (stream-native warmup, 3K steps), and EXP034B (open-gate hardening, 800 steps, matched conditions). The magnitude decreases from EXP025 to EXP034B, likely because EXP034B uses a shorter training budget where all conditions are closer to initialization, but the direction is consistent. This convergent evidence across independent experimental setups strengthens the symbolic-utility claim beyond a single-experiment artifact.

Gate behavior as a diagnostic. The learned gate provides a diagnostic complementary to CE: it is higher for aligned S_{plan} (0.177) than shuffled (0.167) or dummy (0.163) in EXP034B, consistent with the model learning to trust the aligned stream more. The gate does not reach large values (maximum ≈ 0.18), indicating the symbolic stream supplements rather than dominates the acoustic prediction path at 800 steps. Longer training or stronger symbolic supervision (e.g., via a language-model-supervised tokenizer) may open the gate further.

MIR recovery caveats. The EXP031A probing results highlight a critical methodological point: high MIR accuracy is not sufficient evidence for symbolic content if the zero-feature baseline achieves comparable performance. Meter accuracy and instrument micro-F1 are largely explained by prior distributions, and must be excluded from headline claims until label-shuffle controls confirm feature-specific learning.

8 Limitations

No decoded generation evidence. All experiments in this paper evaluate cross-entropy on acoustic token prediction (C_a) under oracle symbolic input. We do not yet have waveform-level evidence that S_{plan} conditioning changes the perceptual quality or structural fidelity of decoded audio. Generating audio from EXP034B checkpoints and comparing aligned versus shuffled outputs perceptually remains future work.

Oracle symbolic input. In all stream experiments, S_{plan} tokens are derived from the ground-truth MIDI file aligned with the audio window. At inference time, aligned MIDI would not be available for arbitrary audio inputs. A practical system requires either audio-to-MIDI transcription or a separate audio-to-S_{plan} predictor, neither of which is trained yet.

No LLM-supervised tokenizer. The current S_{plan} is trained via reconstruction and MIR objectives but not via an LLM decoder objective. A language-model-supervised tokenizer (S_{LLM}) would allow text-language grounding of symbolic tokens, enabling richer downstream tasks such as music captioning and symbolic QA. We treat S_{LLM} as planned future work.

Slakh-only scope. All experiments use Slakh, a synthesized paired audio-MIDI corpus. Generalization to real-world recordings (where aligned MIDI is unavailable) requires either transcription or domain adaptation, which are outside the current scope.

Pending MIR controls. EXP031A label-shuffle and target-prior controls are incomplete. Instrument macro-F1 and several other MIR axes cannot be claimed as robust symbolic evidence until these controls pass.

9 Conclusion

We presented S_{plan}, a compact learned symbolic planning token that compresses MIDI-derived musical structure into a fixed-length RVQ stream for integration into unified audio language models. Through a systematic evaluation ladder with aligned versus shuffled versus dummy controls, we demonstrated that S_{plan} carries musically meaningful content: it reduces acoustic token prediction uncertainty in three independent experiments, opens the symbolic gate more than corrupted baselines, and supports recovery of pitch-class, key, and density attributes under probing. The strongest evidence comes from EXP034B, a complete four-condition matched hardening experiment, where aligned S_{plan} achieves test CE 5.600 versus shuffled 5.614 and reason-only 5.859.

The central claim of this work is conservative but supported: S_{plan} is a compact symbolic codec with acoustic-token utility and music-structure content, as verified by controlled Slakh experiments. We believe the careful experiment design—with matched corruptions at every stage, honest reporting of negative results, and axis-level MIR caveats—provides a methodological template for future symbolic-fusion work in audio language modeling.

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A EXP024 Fixed-Window Gallery

Table 7 shows per-window reconstruction statistics for three validation examples from Slakh Track01501.

Window	Binary Acc	Cont. MSE	Unique codes (books 1–4)
w000 (0–30 s)	1.000	0.00174	15 / 15 / 17 / 17
w001 (30–60 s)	1.000	0.00246	20 / 24 / 26 / 25
w003 (90–120 s)	1.000	0.00130	17 / 17 / 21 / 25

Table 7: EXP024 fixed validation window statistics. All windows achieve perfect binary feature reconstruction.

B EXP029 Fixed-Window Case Spectrum

Table 8 shows per-window loss for four validation examples, including both strong positive cases and boundary cases, to prevent cherry-picking the qualitative evidence.

Window	Aligned	Shuffled	Dummy	Margin
w000	5.226	5.301	5.280	+0.054
w001	6.159	6.179	6.163	+0.004
w002	6.097	6.104	6.091	−0.006
w003	5.876	5.893	5.869	−0.007

Table 8: EXP029 per-window validation losses. Margin is aligned minus best control (negative = boundary case where aligned is not best). Both positive and negative cases are included.

C EXP034B Per-Codebook Cross-Entropy

Table 9 shows per-codebook test CE for EXP034B. The symbolic benefit is distributed across codebooks, with the strongest gains in books 0 and 3 which correspond to lower-frequency content.

Condition	cb0	cb1	cb2	cb3	cb4	cb5	cb6	cb7
Reason-only	5.60	6.00	3.45	5.28	6.02	6.48	6.87	7.16
Aligned S_{plan}	**5.24**	5.71	**3.19**	**4.98**	5.75	6.25	6.68	7.01
Shuffled	5.26	5.73	3.20	4.99	5.76	6.26	6.69	7.02
Dummy	5.26	5.73	3.20	4.99	5.76	6.26	6.69	7.02

Table 9: EXP034B per-codebook test CE. Aligned S_{plan} achieves the lowest CE in books 0, 2, and 3.